

KOCHI METRO RAIL LIMITED

ISO 9001:2015 Certified | CIN: U60100KL2011SGC028710

SAFETY SYSTEMS FITNESS CERTIFICATE

Certificate No: KMRL/SAF/2026/00067

Date of Inspection: 2026-02-19

Train ID:	T-001	Inspector:	R. Kumar
Depot:	Muttom Depot	Valid Until:	2026-08-19

INSPECTION RESULTS - SAFETY PARAMETERS

Param ID	Parameter Name	Unit	Measured Value	Acceptance Range
SAF-001	Fire Detection Response Time	s	18.5	<= 30
SAF-002	Emergency Brake Distance	m	128.0	<= 150
SAF-003	Door Interlock Holding Force	N	620.0	>= 500
SAF-004	Emergency Lighting Duration	min	105.0	>= 90
SAF-005	Smoke Obscuration	%	6.2	<= 10
SAF-006	Deadman Device Response Time	s	2.1	<= 3
SAF-007	Crashworthiness Long. Force	kN	1720.0	>= 1500
SAF-008	Earthing Resistance	Ohm	0.45	<= 1
SAF-009	Emergency Egress Door Width	mm	720.0	>= 650
SAF-010	PSD-Train Door Alignment Tol.	mm	32.0	<= 50

Summary: All safety system parameters for Train T-001 have been tested and are within acceptable limits as per KMRL safety standards and EN 45545 / EN 15227 requirements. Fire detection system activated in 18.5 seconds (threshold: 30s). Emergency braking distance from 80 km/h measured at 128.0 m on level dry rail (limit: 150m). Door interlock force at 620 N, well above the 500 N minimum. Emergency lighting sustained for 105 minutes on battery. Smoke obscuration at 6.2% per metre. Deadman response at 2.1 seconds. Crashworthiness longitudinal force capacity at 1720 kN. Earthing resistance at 0.45 Ohm. Emergency egress door width at 720 mm. PSD alignment tolerance at 32 mm.

This safety systems inspection has been conducted in accordance with KMRL/QA/CERT/2024 standards. All measurements taken using calibrated instruments (NABL accredited).

Inspecting Officer:

R. Kumar
Sr. Rolling Stock Inspector

Authorized Signatory:

S. Menon, Chief Engineer
KMRL Operations